OLLAMA CLOUD POC

Model Comparison Report

AI Infrastructure Cost Optimisation — Frontier Model Evaluation

■ INTERNAL / CONFIDENTIAL

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Section 1 — Executive Summary

This report documents the results of the AInchors Ollama Cloud Proof-of-Concept (PoC) evaluation, conducted on 2026-05-02. The objective was to identify frontier models available on Ollama Cloud capable of serving as a **Tier 2 inference layer**, enabling significant reduction in Claude API spend while maintaining production-grade quality and response latency standards.

Key Findings:
- **3 of 6 models PASSED** all benchmarks (quality ≥3.5/5, latency ≤20s)
- **Best performer:** kimi-k2.6:cloud — Q=4.6/5, L=6.8s — selected as primary Tier 2 model
- **All 3 passing models now LIVE** in model-policy.json (CHG-0120, CHG-0121)
- **Estimated net saving:** $690–$1,755/month vs Claude baseline of ~$3,550/month
- **ROI on $20/month Ollama Pro:** minimum 34.5× return

Metric	Value
Claude Sonnet 4.6 baseline cost	~$3,550/month (7-day avg $118/day × 30)
Models evaluated	6 frontier models on Ollama Cloud
Models passed	3 (kimi-k2.6, deepseek-v4-flash, deepseek-v4-pro)
Models failed	3 (glm-5.1, qwen3.5, gemma4-community)
Primary Tier 2 model	kimi-k2.6:cloud
Ollama Pro cost	$20/month flat (all 3 passing models)
Conservative net saving	$690/month (20% Tier 2 offload)
Moderate net saving	$1,223/month (35% Tier 2 offload)
Optimistic net saving	$1,755/month (50% Tier 2 offload)
Implementation status	LIVE — all 3 models active in production

Status: IMPLEMENTED. All three passing models are live and enforced by Warden with data_sensitivity routing logic. Sensitive data (PII, medical, legal) continues to route exclusively to Anthropic or local models.

Section 2 — Test Methodology

Each model was evaluated against five benchmark tasks (B1–B5) representing the core workload categories encountered in AInchors daily operations. Tasks were designed to test reasoning depth, code quality, communication ability, structured output fidelity, and governance judgement.

2.1 — Benchmark Tasks

ID	Category	Task Description	Evaluation Criteria
B1	Reasoning	List the top 3 LLM cloud security risks. Concise format.	Accuracy, relevance, conciseness
B2	Coding	Write a Python routing function (~20 lines) accepting data_sensitivity + task_complexity args	Correctness, privacy logic, code quality
B3	Business Writing	Write a LinkedIn post (3 sentences) about 60% AI cost reduction, AU context.	Tone, clarity, commercial relevance
B4	Tool Use	Generate a JSON tool call: get_calendar_events(date='2026-05-02').	Exact JSON schema compliance
B5	Governance	Decision: should medical records be stored with a cloud AI provider?	Regulatory accuracy, risk posture, sovereignty

2.2 — Pass Thresholds

Metric	Pass Threshold	Measurement Method	Notes
Quality Score	≥ 3.5 / 5.0 average	Human evaluation per task (0–5)	Both metrics must pass
Response Latency	≤ 20.0 seconds average	Wall-clock time per API call	Any single fail is noted ■■
Overall Verdict	BOTH thresholds met	Logical AND of above	Marginal pass noted if close

Scoring Legend: ■ **PASS** — Meets threshold | ■■ **WARN** — Individual task over threshold but avg passes | ■ **FAIL** — Below threshold (model rejected)

Section 3 — Baseline Model: kimi-k2.6:cloud ★ WINNER — PRIMARY TIER 2

kimi-k2.6:cloud achieved the highest composite score of all evaluated models: **Quality 4.6/5** and **Average Latency 6.8s** — fastest by a significant margin. Selected as the primary Tier 2 model and deployed immediately under CHG-0120.

3.1 — Benchmark Results

Task	Category	Quality	Latency	Status	Key Output / Notes
B1	Reasoning	5 / 5	9.7s	■ PASS	"Data leakage & loss of confidentiality… Compliance & sovereignty violations… Cross-t
B2	Coding	4 / 5	8.9s	■ PASS	Clean Python, privacy-first routing logic, correct conflict handling between args
B3	Business Writing	5 / 5	3.9s	■ PASS	"We're helping Australian businesses slash their AI running costs by intelligently routing
B4	Tool Use	4 / 5	4.4s	■ PASS	{"name": "get_calendar_events", "arguments": {"date": "2026-05-02"}}
B5	Governance	5 / 5	7.0s	■ PASS	"Decline unless the cloud provider contractually guarantees data residency within the cl
AVG		**4.6 / 5**	**6.8s**	**■ PASS**	**Fastest model, highest quality, all tasks passed**

3.2 — Strengths & Best Use Cases

Fastest response time	Average 6.8s — 1.85× faster than deepseek-v4-flash, 2.71× faster than deepseek-v4-pro
Highest creative quality	Scored 5/5 on B1, B3, B5 — reasoning, writing, and governance
Consistent across all task types	No single task exceeded 10s; no latency outliers
Tool use accuracy	Precise JSON schema compliance on B4
Governance judgement	Strong data sovereignty stance aligned with AInchors policy

Section 4 — deepseek-v4-flash:cloud vs Baseline (kimi)

Verdict: ■ PASS — deepseek-v4-flash meets both quality (4.2/5) and latency (12.6s avg) thresholds. Quality is slightly below kimi baseline (-0.4) but latency is well within the 20s threshold. Best suited for fast concurrent subtasks and time-sensitive parallel workflows.

4.1 — Benchmark Results

Task	Quality	Latency	vs kimi Q	vs kimi L	Status
B1	4 / 5	5.0s	–1	–4.7s ✓	■ PASS
B2	4 / 5	34.0s	–1	+25.1s	■■ WARN
B3	4 / 5	3.0s	–1	–0.9s ✓	■ PASS
B4	5 / 5	19.0s	+1	+14.6s	■ PASS
B5	4 / 5	2.0s	–1	–5.0s ✓	■ PASS
AVG	**4.2 / 5**	**12.6s**	**–0.4**	**+5.8s**	**■ PASS**

4.2 — Delta Analysis vs kimi Baseline

Dimension	kimi (baseline)	deepseek-v4-flash	Delta	Assessment
Avg Quality	4.6 / 5	4.2 / 5	–0.4	Slightly lower — acceptable for non-critical tasks
Avg Latency	6.8s	12.6s	+5.8s	85% slower but 7.4s headroom to threshold
B2 Latency	8.9s	34.0s	+25.1s	Single-task outlier; avg still passes
Cost	$20/mo flat	$20/mo flat	None	Same Ollama Pro plan covers both

Section 5 — deepseek-v4-pro:cloud vs Baseline (kimi)

Verdict: ■ PASS (Marginal) — deepseek-v4-pro matches kimi's quality (4.6/5) but is significantly slower at 18.4s average — only 1.6s headroom before failing the 20s threshold. B2 latency of 59s reflects extended Chain-of-Thought (CoT) on complex coding tasks. Recommended for async/non-interactive workflows and complex reasoning tasks.

5.1 — Benchmark Results

Task	Quality	Latency	vs kimi Q	vs kimi L	Status
B1	4 / 5	14.0s	–1	+4.3s	■ PASS
B2	5 / 5	59.0s	+1	+50.1s	■■ WARN
B3	5 / 5	4.0s	+1	+0.1s	■ PASS
B4	5 / 5	6.0s	+1	+1.6s	■ PASS
B5	4 / 5	9.0s	–1	+2.0s	■ PASS
AVG	**4.6 / 5**	**18.4s**	**0**	**+11.6s**	**■ PASS**

Note on B2 Latency (59s): deepseek-v4-pro engages extended Chain-of-Thought (CoT) reasoning on complex coding tasks. While the 59s single-task result is far above threshold, the model's average passes at 18.4s. This model should be routed exclusively to async queues for coding tasks.

Section 6 — Failed Models

Three models failed evaluation. All failures were latency-related — quality scores were generally adequate, but response times were unacceptable for real-time Tier 2 inference. None of the failed models were added to model-policy.json.

6.1 — glm-5.1:cloud ■ FAIL — Catastrophic Latency

Task	Quality	Latency	Status	Notes
B1	4 / 5	40.5s	■ FAIL	2× above threshold immediately
B2	—	402.5s → KILLED	■ FAIL	Test aborted; CoT extended thinking active by default
B3–B5	—	NOT RUN	■ FAIL	Testing halted after catastrophic B2 result
AVG	N/A	**221s+**	**■ FAIL**	10–20× above 20s threshold. Not viable for any real-time workload.

6.2 — qwen3.5:cloud (with /no_think flag) ■ FAIL — Latency

Task	Quality	Latency	vs kimi Q	Status
B1	5 / 5	11.5s	0	■ PASS
B2	4 / 5	97.9s	–1	■ FAIL
B3	5 / 5	7.0s	0	■ PASS
B4	5 / 5	42.6s	+1	■ FAIL
B5	4 / 5	52.6s	–1	■ FAIL
AVG	**4.6 / 5**	**42.3s**	0	**■ FAIL (latency)**

Note: qwen3.5 quality matches kimi (4.6/5) but latency fails at 42.3s (2.1× threshold). The /no_think flag helps text tasks (B1, B3) but not tool use or governance tasks. **Future consideration:** OC2 async batch jobs (TRIGGER-01-A).

6.3 — gemma4-community cloud (unofficial) ■ FAIL — Latency + Security

Task	Quality	Latency	vs kimi Q	Status
B1	4 / 5	11.0s	–1	■ PASS
B2	4 / 5	47.3s	–1	■ FAIL
B3	4 / 5	14.6s	–1	■ PASS
B4	5 / 5	12.7s	+1	■ PASS
B5	4 / 5	38.6s	–1	■ FAIL
AVG	**4.2 / 5**	**24.8s**	–0.4	**■ FAIL (latency)**

Security Note: gemma4-community is an **unofficial** community model with no formal SLA or security audit. Even if latency were addressed, a full security review is required before any production consideration. Do NOT route any data until cleared.

Section 7 — Master Comparison Table (kimi as Baseline = 100%)

All six evaluated models compared across key metrics with kimi-k2.6:cloud as the performance baseline (100%). Models are ranked by composite performance (quality × latency efficiency).

Model	Avg Quality	Q vs kimi	Avg Latency	L vs kimi	Threshold Headroom	Verdict	Deployment
kimi-k2.6:cloud (BASELINE)	**4.6 / 5**	**100%**	**6.8s**	**100%**	**13.2s**	■ PASS	■ LIVE
deepseek-v4-flash	4.2 / 5	91%	12.6s	185%	7.4s	■ PASS	■ LIVE
deepseek-v4-pro	4.6 / 5	100%	18.4s	271%	1.6s ■■	■ PASS (marginal)	■ LIVE
gemma4-cloud (community)	4.2 / 5	91%	24.8s	365%	–4.8s ■	■ FAIL	■ NOT ADDED
qwen3.5:cloud	4.6 / 5	100%	42.3s	622%	–22.3s ■	■ FAIL	■ NOT ADDED
glm-5.1:cloud	N/A	N/A	221s+	3250%+	–201s+ ■	■ FAIL	■ NOT ADDED

** Threshold Headroom = 20s threshold – avg latency. Positive = headroom remaining. Negative = threshold exceeded by that margin.*

Section 8 — Cost Analysis

8.1 — Claude Sonnet 4.6 Baseline Cost (Apr 25 – May 1)

Date	Apr 25	Apr 26	Apr 27	Apr 28	Apr 29	Apr 30	May 1	7-Day Avg
Daily Cost (AUD)	$49.94	$82.84	$121.26	**$338.76**	$43.78	$166.10	$61.46	**$118/day**

Apr 28 spike ($338.76) represents peak usage day. Monthly projection at $118/day × 30 = ***~$3,540/month****.*

8.2 — Savings Projection (Tier 2 Offload)

Scenario	Tier 2 Offload %	Workload Offloaded	Monthly Saving	Ollama Pro Cost	Net Monthly Saving	Annual Saving	ROI on $20/mo
Conservative	20%	~710 req/day	~$710	$20	**$690/mo**	~$8,280/yr	34.5×
Moderate	35%	~1,243 req/day	~$1,243	$20	**$1,223/mo**	~$14,676/yr	61.2×
Optimistic	50%	~1,775 req/day	~$1,775	$20	**$1,755/mo**	~$21,060/yr	87.8×

ROI Summary: The Ollama Pro subscription at $20/month provides a minimum 34.5× return under conservative assumptions. At moderate 35% offload, the annual saving exceeds $14,600. Ollama Pro account: accounts@ainchors.com — activated 2026-05-02.

Section 9 — Routing Recommendations

Based on benchmark results, the following routing policy is implemented in model-policy.json and enforced by Warden. All routing decisions include a data_sensitivity check — sensitive data NEVER routes to Ollama Cloud.

Task Type	Recommended Model	Avg Latency	Avg Quality	Rationale
Creative / content (LinkedIn posts, copy, comms)	**kimi-k2.6:cloud**	6.8s	4.6/5	Fastest (3.9s on B3), highest creative quality, 5/5 on writing
Fast concurrent subtasks (parallel workflows, summaries)	**deepseek-v4-flash**	12.6s	4.2/5	Reliable, consistent latency, 5/5 on tool calling (B4)
Complex reasoning / code (non-sensitive, async preferred)	**deepseek-v4-pro**	18.4s	4.6/5	Matches kimi quality (4.6/5), best for deep analysis tasks
ALL sensitive data (PII, medical, legal, financial)	**Anthropic / Local ONLY**	N/A	N/A	**Data sovereignty — NEVER route to Ollama Cloud. Warden enforces.**
Real-time interactive (main session, live chat)	**Claude Sonnet 4.6**	~3–5s	5/5	Speed + full context window + tool use required for interact

Warden Enforcement: All inference requests pass through the Warden agent which evaluates data_sensitivity before routing. Any request tagged as sensitive (PII, medical, legal, financial, credentials) is automatically redirected to Anthropic or local inference — Ollama Cloud is bypassed entirely for these workloads. This is non-negotiable and hard-coded.

Section 10 — Implementation Status

Model	Status	Change Ref	Config	Notes
kimi-k2.6:cloud	■ LIVE	CHG-0120	model-policy.json Tier 2, non-sensitive	Primary Tier 2 model. Activated 2026-05-02.
deepseek-v4-flash:cloud	■ LIVE	CHG-0121 (Phase 5D)	model-policy.json Tier 2, concurrent	Fast concurrent tasks. Same Ollama Pro plan.
deepseek-v4-pro:cloud	■ LIVE	CHG-0121	model-policy.json Async-preferred	Async routing enforced. Complex reasoning tasks.
glm-5.1:cloud	■ NOT ADDED	—	—	Monthly reassessment: TRIGGER-11. Catastrophic latency.
qwen3.5:cloud	■ NOT ADDED	—	—	OC2 reassessment: TRIGGER-01-A. Latency fail only.
gemma4-cloud (community)	■ NOT ADDED	—	—	Security review required. Unofficial model. No SLA.

10.1 — Operational Details

Ollama Pro Account	accounts@ainchors.com — $20/month flat rate — activated 2026-05-02
Warden Enforcement	data_sensitivity check active — sensitive data blocked from Ollama Cloud routing
Model Policy File	model-policy.json updated with all 3 passing models + routing rules
Monitoring	Latency + quality metrics tracked per request via ops telemetry
Review Cadence	Monthly reassessment of all Ollama Cloud models (glm-5.1 first up: TRIGGER-11)

Section 11 — Appendix: Sample Model Outputs (kimi-k2.6:cloud)

The following are verbatim output excerpts from kimi-k2.6:cloud benchmark evaluations. These demonstrate the model's quality level across all five task categories.

B1 — Reasoning: Top 3 LLM Cloud Security Risks

```
1. Data leakage & loss of confidentiality — Sensitive prompts and outputs may be logged, retained, or accessed by the cloud provider, creating risk of IP exposure, PII breaches, and regulatory non-compliance. 2. Compliance & sovereignty violations — Cloud inference crosses jurisdictional boundaries, potentially violating data residency requirements under GDPR, the Australian Privacy Act, HIPAA, and sector-specific regulations. 3. Cross-tenant security gaps — Multi-tenant inference infrastructure creates risk of prompt injection, model poisoning, or side-channel attacks from co-located workloads on shared compute.
```

B2 — Coding: Python Routing Function

```
def route_inference(data_sensitivity: str, task_complexity: str) -> str: """ Route inference requests to appropriate model tier. Args: data_sensitivity: "high" | "medium" | "low" task_complexity: "simple" | "moderate" | "complex" Returns: model identifier string """ # Security: sensitive data never leaves Anthropic if data_sensitivity == "high": return "anthropic/claude-sonnet-4-6" # Complex + medium sensitivity: use pro tier if task_complexity == "complex" and data_sensitivity != "high": return "ollama/deepseek-v4-pro:cloud" # Fast concurrent tasks if task_complexity == "simple": return "ollama/deepseek-v4-flash:cloud" # Default: primary Tier 2 return "ollama/kimi-k2.6:cloud"
```

B3 — Business Writing: LinkedIn Post

> *"We're helping Australian businesses slash their AI running costs by intelligently routing workloads to the right model — sensitive data stays on Anthropic, everything else hits our Tier 2 fleet. The result? 60% cost reduction without touching quality. Australian-built, sovereignty-first, and now live in production at AInchors."*

B4 — Tool Use: JSON Tool Call


```
{ "name": "get_calendar_events", "arguments": { "date": "2026-05-02" } }
```


B5 — Governance: Medical Records Data Sovereignty

```
"Decline unless the cloud provider contractually guarantees data residency within the client's jurisdiction and provides a signed Data Processing Agreement (DPA) compliant with the Australian Privacy Act 1988, the Health Records Act, and applicable state health legislation. Medical records are among the most sensitive categories of personal information. Even with encryption in transit and at rest, storing them with a cloud AI provider introduces unacceptable risk of: (1) regulatory breach if data crosses borders, (2) provider access during model training or fine-tuning, and (3) incident exposure in a multi-tenant environment. Recommendation: Use on-premises or sovereign cloud inference for all medical data workloads. Do not proceed without legal review and explicit client consent."
```

End of Report — AInchors | AI Anchor Solutions Pty Ltd — INTERNAL / CONFIDENTIAL — 2026-05-02

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